# Ferdinand LACROIX

## Engineering Student – Télécom SudParis (Institut Polytechnique de Paris)

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## Education

**Universidad Carlos III de Madrid (UC3M), Spain** Sept. 2026 – Jan. 2027

*Exchange Semester – Computer Science and Engineering*

- Coursework: Cybersecurity Systems, Cybersecurity and Privacy, High Performance Computing, Data Processing, Digitization of Complex Systems Engineering, Project Management in Informatics Engineering

**Télécom SudParis – Institut Polytechnique de Paris, France** Sept. 2025 – June 2026

*Master's-level Engineering Degree (Grande École programme) – currently in 2nd of 3 years*

- Core curriculum: Networks, Signal & Communication Theory, Electronics, Databases, Software Development
- Member school of IP Paris, alongside École Polytechnique, ENSTA, ENSAE and Télécom Paris
- GATE team project (course): treasurer, managed the budget of a 150-player LAN party

**Lycée Saint-Louis, Paris, France** Sept. 2022 – July 2025

*Preparatory Classes for the Grandes Écoles (PCSI / PSI)*

- One of France's leading preparatory schools for engineering
- Highly selective programme in Mathematics, Physics & Engineering preparing for national entrance exams

**Lycée Français de Barcelone, Barcelona, Spain** July 2022

*French Baccalauréat – Highest Honours*

- Majors in Mathematics and Physics–Chemistry, with the Advanced Mathematics option
- Additional Mathematics and Physics modules in Spanish, preparing for the Spanish university entrance exams

## Experience & Activities

**Data & Tools Intern – Blissim, Paris** June – August 2026

*Beauty subscription e-commerce*

- Migrated 20+ dashboards from Periscope to Metabase, rewriting SQL queries and reorganising them by team.
- Ran the daily allocation of personalised monthly boxes for new subscribers across 5 countries.
- Built CHOIPO, a full-stack internal app (React, TypeScript, Express, BigQuery, PostgreSQL) replacing a manual Excel workflow, reducing errors and laying the groundwork for automated allocation.

**Tennis Coach – Children aged 5–10** Summer 2022

*UAS Saint-Cloud, Saint-Cloud*

- Led group sessions independently, focusing on mental skills: frustration, concentration, perseverance.

**Student Associations** 2025 – Present

*Télécom SudParis*

- Vice-President, Déclic (Photography Club): event coverage, photo editing and workshop organisation.
- Logistics & Events, Xtreme (Outdoor Club): co-organising the Bike & Run race for 5 IP Paris schools.
- Kryptosphere (AI, Blockchain & DeFi Club): attending weekly training sessions, incl. machine learning.

## Projects

**Software & Web Development** 2024 – Present

*Academic and Independent Projects*

- *FIRE!* (Java, JavaFX): strategy game with a Monte Carlo Tree Search AI (academic project).
- Web apps: RacketLab, catalogue of top 100 ATP/WTA players' rackets with a 3D viewer (React, TypeScript, three.js); Mysteria, riddle platform (HTML/CSS/JavaScript); TCG booster app (Next.js).

**Research Projects (TIPE)** 2023 – 2025

*Supervised research projects assessed at the national entrance exams*

- Helmholtz Resonators: modelled and experimentally measured low-frequency sound absorption.
- Badminton Racket: modelled energy transfer to the shuttle, treating the racket as a harmonic oscillator.

## Skills, Languages & Interests

### Technical

*Languages:* Python, Java, SQL, TypeScript/JavaScript, Bash

*Frameworks:* NumPy, XGBoost, JavaFX, React, Next.js, Express

*Tools:* Git, LaTeX, BigQuery, PostgreSQL, Metabase, Node.js

### Languages

**French** – Native

**English** – C1–C2 (fluent)

**Spanish** – Bilingual

*9 years of cultural immersion in Latin America & Spain*

### Interests

**Sports:** Tennis (French ranking 15/2), volleyball, skiing, running (half-marathon, marathon)

**Arts:** Photography (Lightroom), guitar